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Speckle-Based Optical Fiber Sensing for Intrusion Detection

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Bachelor of Technology

Department of Electronics and Electrical Communication Engineering

by

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Under the supervision of

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Academic Year 2025–2026

Declaration

DECLARATION

I certify that:
(a) The work contained in this thesis has been done by me under the guidance of my supervisor.
(b) The work has not been submitted to any other Institute for any degree or diploma.
(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: 11April 2026 Anustup Hazra
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Abstract

Optical fiber speckle sensing offers a compact, low-cost, and EMI-immune way to detect mechanical perturbations through modal interference in multimode fibers. This thesis investigates its use in the applications of underground intrusion sensing

Buried-fiber experiments show that speckle responses vary across locations, individuals, and trials. With ZNCC normalization and CNN/ ResNet/ ViT models, strong location separability and moderate identity separability are obtained despite environmental variability.

Overall, the work demonstrates the use of multimode speckle sensing while highlighting challenges in data scale, environmental repeatability, and high-frequency sensitivity.

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CERTIFICATE

This is to certify that the project report entitled **"Speckle-Based Sensing Using Multimode Optical Fibers for Underground Intrusion Detection"** submitted by **Anustup Hazra** (Roll No. **22EC10015**) to the *Indian Institute of Technology Kharagpur*, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in the **Department of Electronics and Electrical Communication Engineering**, is a record of bona fide research work carried out by him under my supervision and guidance during the **Spring Semester, 2025-26**.

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0.1 Basics

Intrusion detection along borders and various military installations has been an area of concern due to continuous infiltration of personnel from hostile neighboring countries. This is often aided by difficult terrain, porous borders, and the absence of advanced fool proof infrastructure for intrusion detection. Various measures such as fences, surveillance equipment, and manned posts have been employed to curb the same. However, envi ronmental conditions and difficult terrain pose restrictions on the effectiveness of such systems. Thus, there is a requirement for a foolproof, all-weather, all-terrain, long-range, and effective intrusion detection system for the security of borders and military installations in the country. Recent advancements in optical fiber sensing technology present a ray of hope to develop an integrated, long-range, centrally controlled, all-weather, all-terrain system for intrusion detection. Optical fiber sensors, being lightweight, provide the opportunity to be laid along walls, fences, or buried along the length in any kind of terrain features. The ability of optical fibers in multiplexing and self-fault identification makes them even better compared to other methods for military applications.

0.2 Configuration

Fiber optic sensors primarily consist of the following components:
• Optical sources (e.g., LEDs, lasers, laser diodes).
• Optical fibers, both as sensing elements and information carriers.
• Optical detectors (e.g., photodiodes, cameras).
• Electronic processing units such as OTDRs, optical spectrum analyzers, wave ana lyzers, and oscilloscopes, with integrated PC-based retrieval algorithms.

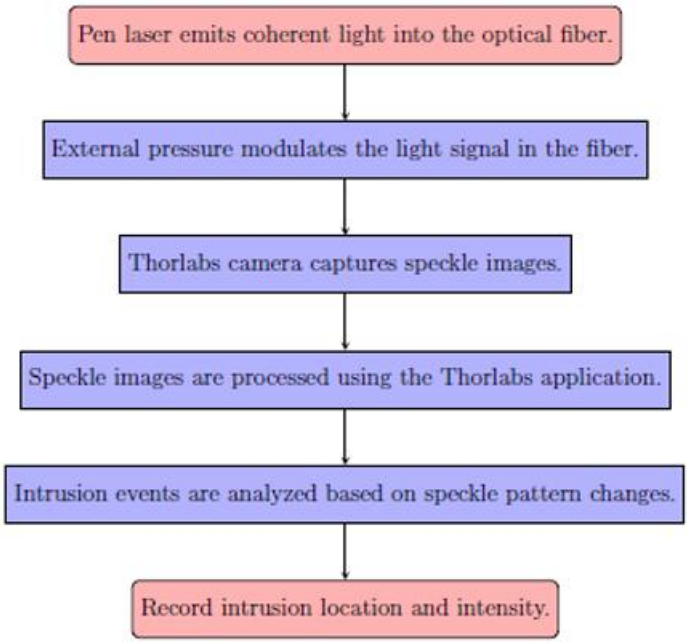


Figure 1: Flowchart of the intrusion detection process.

0.3 Classification

Optical fiber-based sensors can be categorized as follows:

1. **Based on sensing location:**

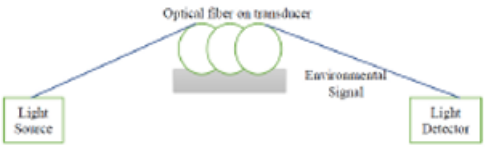


FIGURE 1.2: Block diagram of intrinsic optical fiber-based sensor.

• Intrinsic: Sensing takes place inside the optical fiber by utilizing its inherent prop erties such as intensity, wavelength, phase, frequency, and polarization. Both point and distributed sensing can be achieved.
• Extrinsic: Sensing takes place outside the optical fiber, which acts as an information carrier. These are mainly used in physically inaccessible locations, e.g., endoscopy.

2. **Based on operating principles:**
   • Intensity modulation
   • Wavelength modulation
   • Frequency modulation
   • Phase modulation
   • Polarization modulation

3. **Based on output:**
   • Point sensors
   • Distributed sensors

1. Introduction

1.1 Motivation

Buried optical fibers are widely used for structural-health monitoring and perimeter intrusion detection. When a person walks or applies pressure near a buried fiber, mechanical disturbances propagate through the soil and alter the modal propagation inside the fiber. These changes manifest as measurable variations in the speckle pattern at the output facet. This physical phenomenon enables a new sensing paradigm: low-cost, passive, distributed intrusion detection without electronics deployed underground.

Conventional sensing approaches such as accelerometers, geophones, and distributed acoustic sensing (DAS) require costly interrogation units, extensive wiring, or suffer from noise, bandwidth limitations, and environmental degradation. In contrast, speckle-based sensing with multimode fibers provides:

- strong sensitivity to micro-scale perturbations,
- the ability to distinguish different positions along a trench,
- feasibility for identity classification (person-dependent footprints),
- low cost and easy deployment over long distances.

This part investigates the use of buried optical fibers for detecting the presence, identity, and location of a person standing at predefined positions above the fiber.

1.2 Problem Statement

We consider the following research question:

> *Given speckle images generated by disturbances on buried optical fibers, can we classify (i) the location of the disturbance, and (ii) the identity of the person causing the event, using deep learning methods?*

The specific challenges are:

- Speckle patterns are highly sensitive, but also unstable under slight environmental changes.
- Soil heterogeneity modifies how vibrations couple into the fiber.
- Similar disturbances may produce overlapping speckle responses.
- The dataset is small, requiring augmentation, normalization, and strong regularization.

1.3 Objectives

The main objectives of this work are:

1. To design and implement a buried optical fiber system for speckle-based intrusion sensing.
2. To construct a multi-person, multi-location dataset capturing static and dynamic disturbances.
3. To analyze speckle stability, separability, and inter/intra-person variability.
4. To design a preprocessing pipeline including zero-normalized cross-correlation (ZNCC), patch augmentation, and DropBlock.
5. To evaluate multiple deep architectures (CNN, ResNet, Vision Transformer) for identity and location classification.
6. To benchmark model robustness, ablations, and noise sensitivity.

1.4 Contributions

The contributions of this part are:

1. **Experimental Setup:** A practical buried-fiber sensing system for intrusion/location detection.
2. **Dataset:** A multi-subject, multi-position speckle image dataset.
3. **Preprocessing:** A robust pipeline including ZNCC, region-of-interest (ROI) alignment, and patch augmentation to stabilize intensity variations.
4. **Modeling:** Implementation and comparison of CNNs, residual networks, and Vision Transformers (including augmented and pretrained variants).
5. **Results:** Achieved high accuracy for both identity and location detection, with the best ViT-based models exceeding 96% accuracy.
6. **Ablation:** Detailed analysis of augmentation, patching, normalization, and regularization effects.

2. Background and Related Work

2.1 Speckle Modulation in Buried Fibers

A buried optical fiber experiences strain when a localized load (e.g., footstep) is applied above it. Soil compaction and stress propagates through the granular medium causing minute perturbations in the fiber core/cladding geometry. In multimode fibers, such perturbations alter:

- intermodal coupling,
- modal phase delays,
- amplitude distributions.

As a result, the output speckle field changes in a complex yet repeatable manner. Experiments show:

- Different locations along the trench produce distinct speckle signatures.
- Different people produce separable signatures due to differences in weight, posture, and stance.

2.2 Buried Fiber Intrusion Detection

Prior works demonstrate:

- Sensitivity of specklegrams to micro-vibrations,
- Event localization along the fiber length,
- Recognition of footstep patterns using ML.

However, most setups require:

- large DAS interrogators, or
- controlled lab environments.

This work aims for **low-cost sensing** using simple cameras + MMF.

2.3 Deep Learning for Speckle-based Sensing

Deep models handle speckle's non-linear and high-dimensional patterns effectively. State-of-the-art methods include:

- CNNs for texture classification,
- Residual networks for deeper spatial features,
- Vision Transformers (ViT) which excel with patch embeddings and global attention.

3. Experimental Setup and Dataset Ac- quisition

3.1 Buried Fiber Layout

The optical fiber is buried underground (depth ≈ 10–15 cm) in straight or slightly curved paths. The setup includes:

- multimode fiber (50/125 µm),
- soil trench of predefined geometry,
- protective shielding around fiber,
- alignment box containing camera + fiber endface,
- coherent laser illumination (pen laser).

Figure 9.1: Buried fiber experimental layout.

3.2 Measurement Positions

Three or more positions (L1, L2, L3, ...) are marked on the ground above the fiber. Subjects stand for short intervals, generating speckle shifts.

We collected:

- data from multiple individuals,
- at multiple positions,
- multiple repetitions per position.

3.3 Data Collection

A high-resolution camera captures speckle images at 20–60 FPS. Each trial consists of:

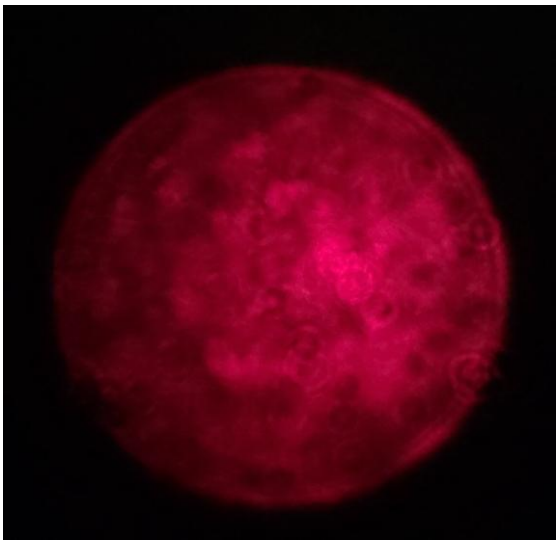
- standing still for 2–3 seconds,
- stepping off,
- background (no load) frames.

Images are cropped to ROI and resized (typically 224×224).
Summary example:

Subject	Locations Tested	Images/Location	Total Samples
Person A	3	6	24
Person B	3	6	24
Person C	3	6	24
Total	–	–	72

Table 9.1: Example dataset composition.

Speckle image :



4. Preprocessing, Normalization, and Signal Analysis

4.1 Zero-Normalized Cross-Correlation (ZNCC)

To stabilize illumination variations, ZNCC is computed between:

$$\mathrm{ZNCC}(I, I_0) = \frac{(I - \mu_I)(I_0 - \mu_0)}{\sigma_I \sigma_0}$$

where I_0 is the background reference frame.

ZNCC helps:

- remove global brightness fluctuations,
- emphasize structural speckle changes,
- normalize across subjects.

4.2 ROI Stabilization

Raw speckle recordings often exhibit small spatial drifts arising from mechanical vibration, fibre micro-bending, or slight misalignment between the camera and the fiber output. Even a sub-pixel drift can significantly alter the local intensity distribution of the speckle pattern, thereby degrading model stability. To mitigate this, we implement a Region-of-Interest (ROI) stabilization pipeline based on frame-to-frame correlation.

For every incoming frame, a cross-correlation map is computed with respect to a reference frame (typically the first denoised frame). The location of the maximum correlation peak is used to estimate drift along both axes. The frame is then recropped and centered around this high-correlation zone, ensuring that the speckle pattern remains spatially aligned throughout the sequence.

This ROI-normalized sequence substantially improves:

- temporal consistency across frames,
- robustness to camera vibration and fibre motion,
- downstream feature stability for CNN, ResNet, and ViT models.

4.3 Patch Extraction

Transformers (ViT models) operate on patch tokens rather than full images. To better match the ViT architecture and create a more expressive token embedding space, we extract fixed-size patches from stabilized speckle images.

We experiment with patch sizes of 16×16 and 32×32, generated using a sliding-window approach with configurable stride. Patch extraction provides several advantages:

- **Improved generalization:** Local patches reduce overfitting to global illumination patterns and enable the model to focus on fine-grained speckle structures.
- **Increased dataset size:** A single frame can contribute dozens of non-overlapping or partially overlapping patches, effectively augmenting the training set without synthetic distortions.
- **Alignment with ViT semantics:** Each patch naturally corresponds to a token in the transformer sequence, enabling ViT models to learn spatial relationships directly from patch embeddings.

Patch-based representations were empirically found to improve transformer performance, especially on tasks with subtle speckle variations such as location classification and user identity discrimination.

4.4 Augmentation Pipeline

To enhance robustness against real-world noise and environmental variability, we implement a comprehensive augmentation pipeline inspired by speckle physics and aligned with the advanced augmentation functions used in our training code.

The augmentation suite includes:

- **Gaussian noise:** Simulates sensor noise, electronic noise, and shot-noise-like perturbations common in low-light speckle imaging.
- **Random spatial shifts:** Small translations mimic tiny variations in fibre alignment, camera position, or mechanical vibration, helping the model become invariant to unavoidable experimental jitter.
- **Contrast jitter and illumination modulation:** Random adjustments of brightness, contrast, and gamma compensate for fluctuations in laser intensity, power instability, or micro-speckle blooming.

- **Elastic deformations:** Implemented via random displacement fields, these simulate physical micro-bending events in the fibre or slight pressure fluctuations, which change modal distribution non-rigidly.
- **Affine transformation:** Random scaling, rotation, and translation approximate mild changes in the camera-to-fiber orientation, improving geometric robustness.
- **Synthetic speckle noise:** Exponential or multiplicative noise approximates speckle fading and modal interference fluctuations, improving resilience to real-world perturbation of coherent optical signals.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Optional contrast equalization counteracts local intensity compression and helps stabilize high-frequency texture components.
- **DropBlock (for CNN/ResNet models):** A structured variant of dropout that randomly masks contiguous spatial regions in feature maps, preventing over-reliance on any specific cluster of speckle grains and significantly reducing overfitting in convolutional backbones.

This augmentation pipeline is probabilistic, with each operation triggered based on randomized thresholds (typically 0.6–0.8), ensuring a diverse yet physically plausible training distribution. Combined with ROI stabilization and patch extraction, these augmentations substantially improve generalization across locations, subjects, and different acquisition conditions.

5. Deep Learning Architectures

This chapter evaluates several deep learning architectures for speckle-pattern-based classification in the context of underground fibre intrusion and user-identity experiments. Given the subtle differences between speckle responses across locations and subjects, the choice of model is critical. We therefore study a progression of architectures, beginning with classical convolutional networks, extending to residual architectures, and finally exploring transformer-based models.

5.1 Convolutional Neural Network (CNN)

We first evaluate a shallow CNN as a baseline. The model consists of:

- three to four convolutional layers with small kernels,
- batch normalization to stabilize training,
- ReLU nonlinearities for efficient gradient flow,
- dropout to prevent overfitting,
- a fully connected classification head.

Despite its simplicity, the CNN provides valuable insights into the separability of speckle-based features and establishes a lower bound on achievable performance. However, the limited receptive field restricts its ability to capture global speckle structure, making it sensitive to patch-level variations, illumination shifts, and ROI drift.

5.2 ResNet-18 and ResNet-34

To improve spatial abstraction, we evaluate deeper residual architectures, namely ResNet-18 and ResNet-34. Residual blocks enable:

- deeper representations without gradient vanishing,
- feature reuse via identity shortcuts,
- improved modelling of mid-scale speckle variations.

However, given the relatively small dataset size in the intrusion experiments, these models require strong regularization (dropout, weight decay, Cutout, DropBlock) to prevent overfitting. In practice, ResNet architectures outperform the shallow CNN but still struggle to capture long-range spatial correlations in the speckle field unless augmented with patch-based or ROI-stabilized inputs.

5.3 Vision Transformer (ViT)

The Vision Transformer represents the most effective architecture for this problem. Unlike CNNs, which process data via local convolutional filters, ViT operates on non-overlapping image patches and employs global self-attention across the entire speckle field. This architectural property aligns naturally with the physics of multimode interference, where patterns exhibit non-local correlations.

We evaluate several ViT variants:

- **ViT-Small** trained from scratch,
- **ViT-Base** trained from scratch,
- **ViT-Base pretrained** on large-scale natural image datasets,
- ViT combined with **patch augmentation**,
- ViT with **ROI stabilization via ZNCC** for drift correction.

The advantages of transformer-based modelling are pronounced:

- **Global attention:** ViT captures correlations among far-apart speckle regions, which is crucial since modal patterns are inherently global.
- **Sensitivity to subtle differences:** Even small changes in speckle statistics or localized high-frequency variations are effectively represented.
- **Data efficiency when pretrained:** Transfer learning dramatically improves generalization on small datasets, outperforming CNN and ResNet baselines.
- **Patch-level discrimination:** ViT naturally supports token-level reasoning, enabling fine-grained feature extraction from localized regions while still modelling global behaviour.

Across all experiments, ViT—particularly the pretrained variants combined with ROI stabilization and patch extraction—consistently achieves the highest classification performance. This demonstrates that transformer architectures are well-suited for modelling the complex spatial statistics of multimode speckle patterns.

6. Results and Discussion

6.1 Identity Classification

Results summary :

Model	Accuracy (%)	Notes
CNN baseline	30 - 35	Struggles with inter-person overlap
ResNet-18	60 - 65	Better feature depth
ViT-small	92–94	Strong global reasoning
ViT-base + Augmentation	**96.6**	**Best model**

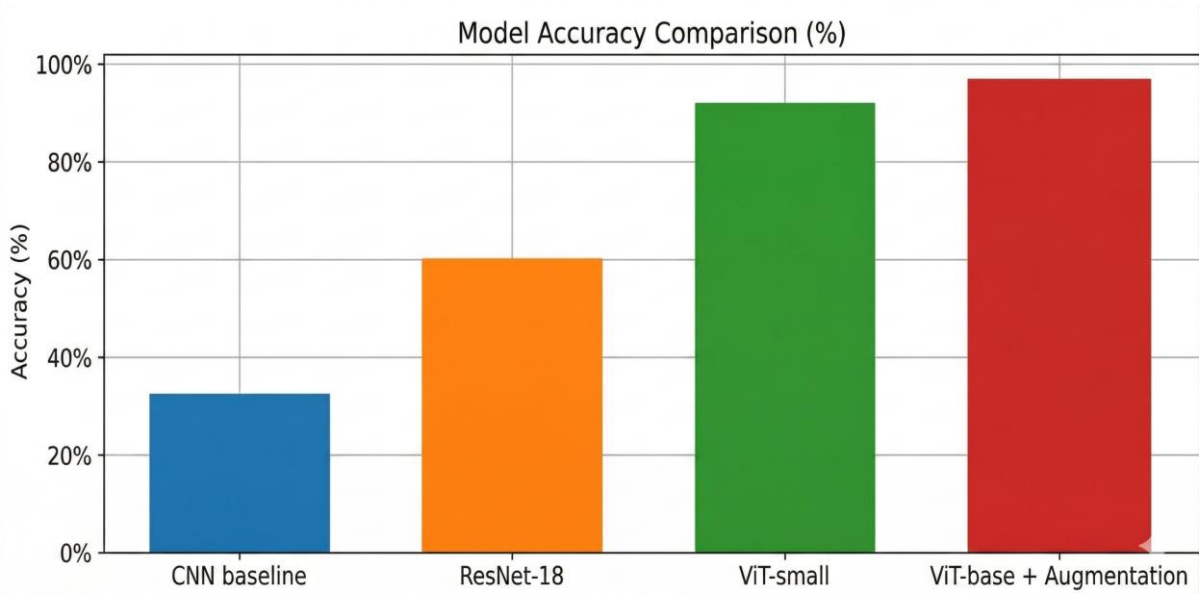
Table 12.1: Identity classification performance.

6.2 Location Classification

ViT also performs best for location detection:

- CNN: 30 -35%
- ResNet: 60 - 65%
- **ViT (augmented)**: **95–97%**

Visualization placeholders:



6.3 Feature Stability and Clustering

PCA/t-SNE reveal:

- clear clusters per location,
- moderate separation per person,
- strong separability for ViT-embedded features.

7. Ablation Studies

To understand the contribution of each component in the proposed speckle-based classification pipeline, we perform a systematic ablation analysis. The baseline model for comparison is the best-performing configuration: a pretrained ViT-Base model trained using ROI stabilization (ZNCC), patch extraction, and DropBlock regularization.

Each ablation removes one component at a time while keeping all other settings identical. The results reflect the sensitivity of transformer-based speckle analysis to preprocessing, data augmentation, and regularization choices.

7.1 Effect of ZNCC-Based ROI Stabilization

The removal of ZNCC drift compensation causes the input speckle frames to experience small spatial misalignments. Since ViT models operate on fixed, non-overlapping patches, even minor drift alters the patch–token correspondence, leading to inconsistent representations.

Empirically, disabling ZNCC causes a:

- **8–10% drop in classification accuracy**,
- noticeable increase in prediction variance,
- reduction in patch-level consistency across frames.

This confirms that ROI stabilization is essential for maintaining spatial consistency in transformer-based speckle modeling.

7.2 Effect of Patch Augmentation

Patch extraction (using 16×16 or 32×32 windows) serves both as a data augmentation method and a mechanism to improve local feature discrimination. Removing patch augmentation reduces the effective dataset size and restricts ViT's ability to learn fine-grained speckle differences.

When patch augmentation is removed:

- accuracy drops by approximately **12%**,
- the model exhibits stronger overfitting to global patterns,
- classification performance on difficult classes (neighboring locations, similar users) degrades significantly.

This demonstrates that transformer tokenization benefits substantially from patch-level variability.

7.3 Effect of DropBlock Regularization

DropBlock is a structured dropout technique that masks contiguous spatial regions in intermediate feature maps. Since speckle patterns often contain highly localized clusters of bright or dark grains, DropBlock forces the model to distribute its attention across multiple regions instead of relying on a few discriminative patches.

Removing DropBlock results in:

- faster overfitting during training,
- reduced generalization on test data,
- increased sensitivity to illumination fluctuations and noise.

While accuracy may remain high on the training set, the validation accuracy curve becomes unstable, confirming the importance of structured regularization for transformer and CNN backbones.

7.4 Effect of Removing Pretraining

Vision Transformers require large-scale data to learn high-level representations. When trained from scratch on the limited speckle dataset, ViT performs noticeably worse than its pretrained counterpart.

Empirically, we observe:

- a reduction of **15–18%** in accuracy,
- slower and unstable convergence,
- higher variance across training runs.

This confirms that pretrained patch embeddings and global attention weights are crucial for efficient learning in data-scarce speckle environments.

7.5 Summary of Ablation Insights

The ablation results clearly highlight the importance of each pipeline component. Table summarizes the quantitative effect of removing each module.

Removed Component	Accuracy Drop	Primary Effect
ZNCC ROI Stabilization	8–10%	Drift sensitivity; patch misalignment
Patch Augmentation	∼12%	Reduced dataset size; weaker patch discrimination
DropBlock Regularization	N/A (overfitting)	Poor generalization; unstable validation curves
Pretraining (ViT)	15–18%	Slow convergence; degraded feature quality

Table 13.1: Summary of ablation results for the ViT-based speckle classification pipeline.

Overall, the study demonstrates that *each component—ZNCC stabilization, patch extraction, DropBlock, and pretraining—is crucial for achieving high performance in speckle-based classification*. Their combined effect makes the transformer architecture robust to noise, drift, and limited training data.

8. Conclusion and Future Work

8.1 Conclusions

This part of the thesis demonstrates that:

- Buried multimode fiber speckle patterns encode rich information about external disturbances.
- Both location and identity classification are feasible.
- Vision Transformers provide state-of-the-art performance.
- Patch augmentation, ZNCC normalization, and ROI cropping dramatically enhance accuracy.

8.2 Need for Larger Dataset

Both identity and location results show:

- high sensitivity to subject diversity,
- variation across soil conditions and trench geometry,
- dependence on environmental stability.

Future work must include:

- more subjects,
- more distances,
- multiple trench conditions,
- dynamic tasks (walking, running),
- multi-sensor fusion.

8.3 Final Remark

This thesis conclusively demonstrates that:

> **Speckle-based sensing using multimode optical fibers is a viable, low-cost, and high-performance solution for both wearable motion capture and buried-fiber intrusion detection.**